CSC343H5 · Introduction to Databases

Sprint 5: Transactions & Concurrency

Released: End of Week 10 (July 28th) **Due:** End of Week 12, Aug 9th at 5 PM ET
Weight: 8% of final grade **Resubmission:** One resubmission within 1 week of grade release

1. Overview

You designed the Research Network schema in Sprint 2, queried it in Sprint 3, and normalised it in Sprint 4. All three assumed you were the only person modifying the database. In this sprint, we'll look at the theory which protects your database even when multiple people work on it.

Everything here is written work: there is no code to run and nothing to submit but `s5.tex` and `s5.pdf`. Questions 1–3 are analysis of given schedules. Questions 4 and 5 apply the same reasoning to your own schema.

Notation, as in lecture: `R1(A)` is transaction 1 reading A, `W2(B)` is transaction 2 writing B, `C1` and `A1` are commit and abort, and `X1(A)`, `S1(A)`, `U1(A)` are exclusive lock, shared lock, and unlock.

Question 1 Precedence Graphs and Serializability (25 marks)

S1: `R1(A) R2(B) W1(A) R3(C) W2(B) R1(B) W3(C) W1(B)`
S2: `R1(A) W2(A) R2(B) W3(B) R3(C) W1(C)`

For **each** schedule, give:

1. a table of every conflicting pair, in schedule order, each labelled **WR**, **RW**, or **WW**;
2. the precedence graph, with every edge labelled by its object;
3. the verdict. If conflict serializable, give an equivalent serial order and state whether that order is unique. If not, name the cycle and say in one sentence why no serial order can exist.

The two answers do not have the same shape. One of these schedules contains a transaction that conflicts with nothing at all.

Question 2 Aborts and Recoverability (20 marks)

S3: `W1(A) R2(A) W2(B) C1 C2` **S4:** `W1(A) R2(A) W2(B) C2 C1` **S5:** `W1(A) W2(A) C1 C2`
S6: `W1(A) R2(A) W2(B) R3(B) W3(C) A1`

1. Complete this table. In the last column, name the single operation pair that decides the row, e.g. "`W1(A)` then `R2(A)`".

Schedule	**Recoverable?**	**ACA?**	**Strict?**	**Deciding pair**
S3				
S4				
S5				

2. **S5** contains no dirty read at all and still fails one of the three properties. Which one, and why does that failure make undo harder? (Two sentences.)

3. **S6** ends in an abort. List every transaction that must be rolled back, in order, with a reason for each. Then state the one property from part (i) that, if enforced, would have made this cascade impossible.

Question 3 Locking and Two-Phase Locking (25 marks)

Schedule **S7** is fully lock-annotated. Both transactions take a lock before every access and release every lock they take.

```
X1(A) R1(A) W1(A) U1(A) X2(A) R2(A) W2(A) X2(B) R2(B) W2(B) U2(A) U2(B) X1(B) R1(B)
W1(B) U1(B) C1 C2
```

1. Every lock in **S7** is legally held, and yet the schedule is not serializable. Show this: give the precedence graph and name the cycle.

2. For T1 and T2 *separately*, state whether the transaction satisfies **2PL**, **Strict 2PL**, both, or neither. Name the exact operation that decides each verdict.

3. Rewrite **S7** so that both transactions obey Strict 2PL, as a two-column table (one column per transaction). Mark every point at which a transaction is denied a lock and must wait, and the operation that later releases it. State the equivalent serial order of your rewrite.

4. Strict 2PL guarantees serializability but not freedom from deadlock. In two sentences, describe a situation over two objects in which two Strict 2PL transactions deadlock, and state what the DBMS does about it.

Question 4 Choosing an Isolation Level (12 marks)

For each of the following operations on *your* Sprint 2 Research Network schema, state the **weakest** SQL isolation level that is safe, and justify it in one sentence. Name the anomaly you are defending against, or say why none applies.

1. Add one Researcher to one Project (a single insert into your project–researcher table).
2. Move a Researcher from one Project to another.
3. Award a new Grant to whichever Project currently has the smallest total allocation.
4. Produce a report of the number of Publications per Project, reading several tables and writing nothing.

Question 5 A Read–Decide–Write Hazard (18 marks)

Sprint 2 modelled Equipment that a Lab owns and may share. Suppose Labs also book time on shared equipment, using a counter of remaining slots and a booking table. The booking transaction is:

```
BEGIN;
SELECT slots_remaining FROM equipment_slot WHERE equipment_id = 7;
-- application checks the value is > 0, and computes the new value
   itself
UPDATE equipment_slot SET slots_remaining = 0 WHERE equipment_id = 7;
INSERT INTO booking (equipment_id, lab_name) VALUES (7, 'Ecology Lab');
COMMIT;
```

The row starts at `slots_remaining = 1`. Two labs run this transaction at the same time, at `READ COMMITTED`.

1. Give an interleaving of the two transactions, in schedule notation over the single object `S` (the `equipment_slot` row), that leaves *two* bookings in the database. List its conflicting pairs and their types, and state whether the schedule is conflict serializable.
2. State exactly which invariant of the data is violated, and which one is not. Be precise: the counter itself never goes negative.
3. Name the isolation level that prevents this. Under that level the second transaction is not blocked, it fails, so say what the application must then do and what it must *not* reuse when it does.
4. Suppose you keep `READ COMMITTED` instead, and change the read to `SELECT ... FOR UPDATE`. Which lock mode from lecture is that, why is the shared-mode version insufficient, and at which step does the second transaction now wait?
5. This one row is touched by every booking request in the system. Name one of the three levers from lecture that would reduce contention on it, describe the concrete change, and state its cost.

2. Submission & Policies

Submit to MarkUs by **Aug 9th, 5:00 PM ET**: `s5.tex` and `s5.pdf`. Your full name, student number, and UTORid must appear in both. All answers must be typed in LaTeX; precedence graphs may be drawn by any means, including by hand and photographed, provided they are legible and embedded in `s5.pdf`.

Resubmission. One resubmission within one week of grade release, with a changelog.

AI usage. Permitted, declaration mandatory: state the tool, what you asked it, and what you understood and applied. Language models are unreliable on precedence graphs and on which anomalies a given isolation level actually permits, and both are graded on the justification rather than the label, so verify anything a tool hands you.